# Insurance AI Assistant Fine-Tuning Project

## Contents

# 1. Project Title

Insurance AI Assistant Using QLoRA, Instruction Fine-Tuning, and DPO Alignment.

# 2. Domain Selected

**Commercial Insurance & Underwriting**

This project focuses on building an AI-powered insurance assistant capable of answering underwriting, risk assessment, policy evaluation, and insurance-related business questions.

The objective was to adapt a foundation Large Language Model (LLM) to understand insurance-specific terminology, underwriting guidelines, risk management concepts, and respond professionally to users.

# 3. Business Problem

Insurance professionals spend significant time reviewing submissions, analyzing risks, interpreting policy requirements, and answering repetitive underwriting questions.

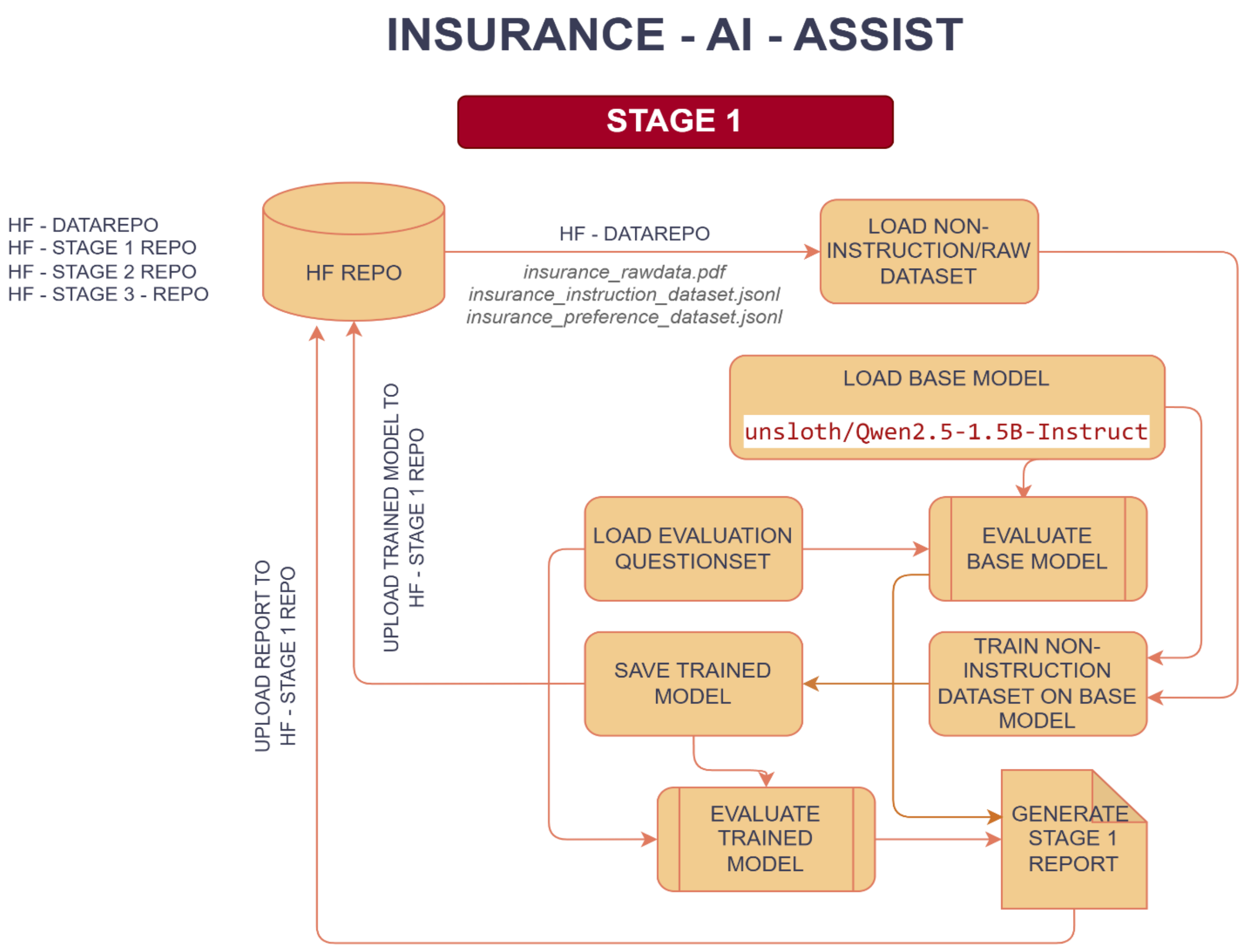
Challenges include:

- Manual review of insurance submissions
- Inconsistent underwriting responses
- Slow turnaround time
- Knowledge dependency on experienced underwriters
- Limited accessibility to insurance expertise

The goal of this project is to develop an Insurance AI Assistant capable of:

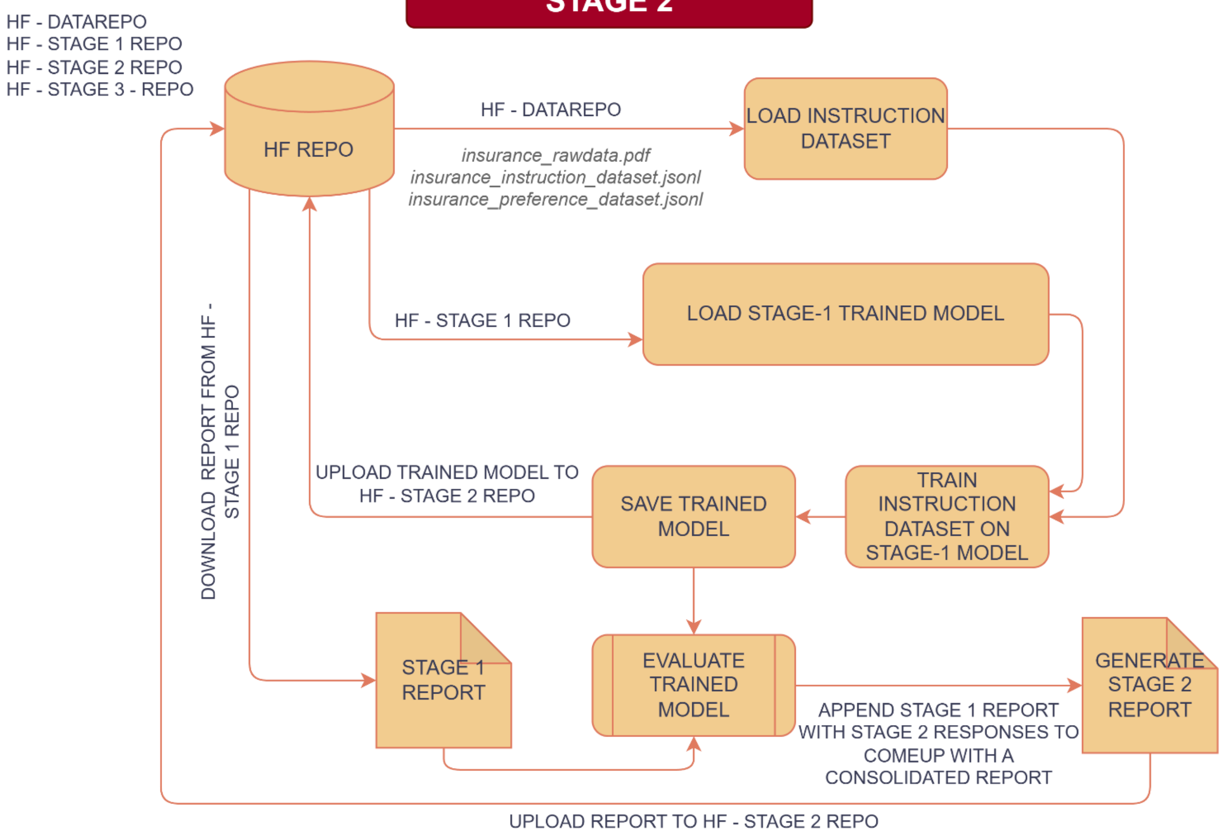
- Understanding insurance terminology
- Providing underwriting guidance
- Assisting with risk evaluation
- Answering insurance-related questions
- Producing professional and consistent responses

# 4. Architecture



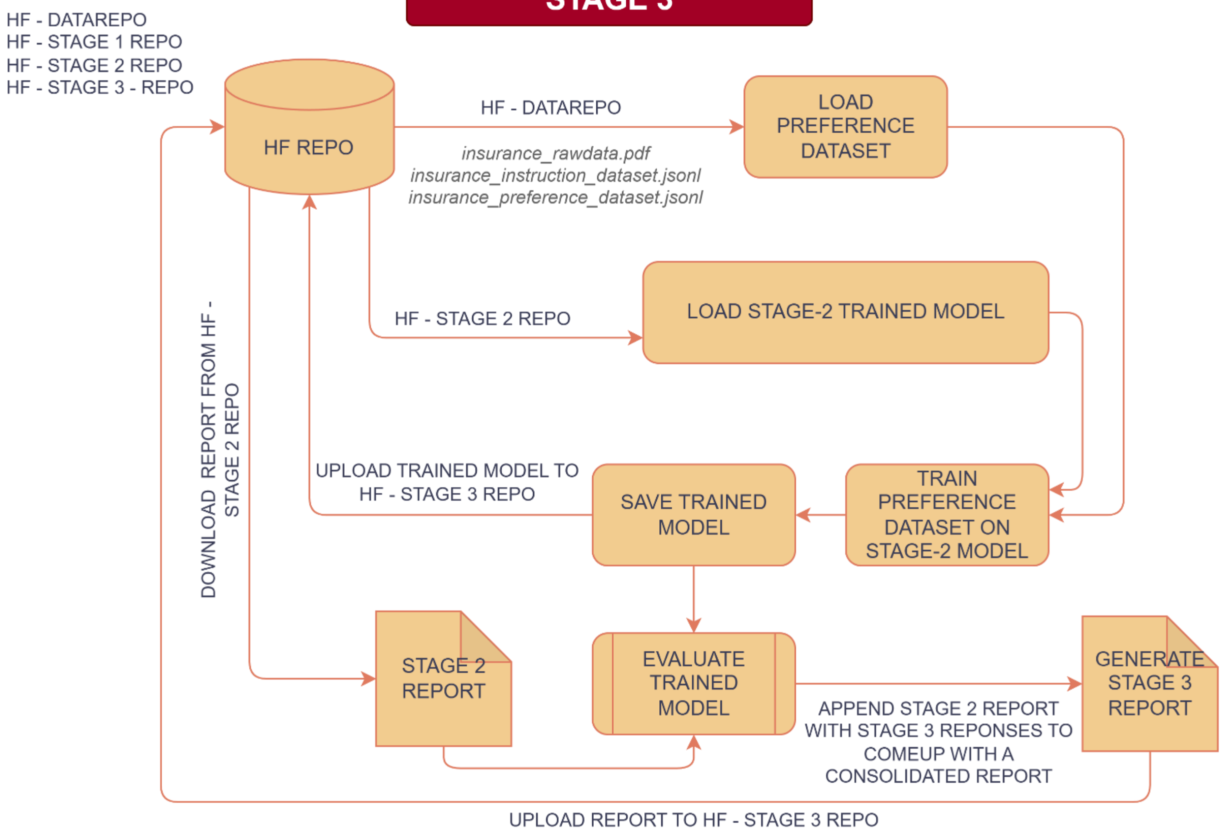
# INSURANCE - AI - ASSIST

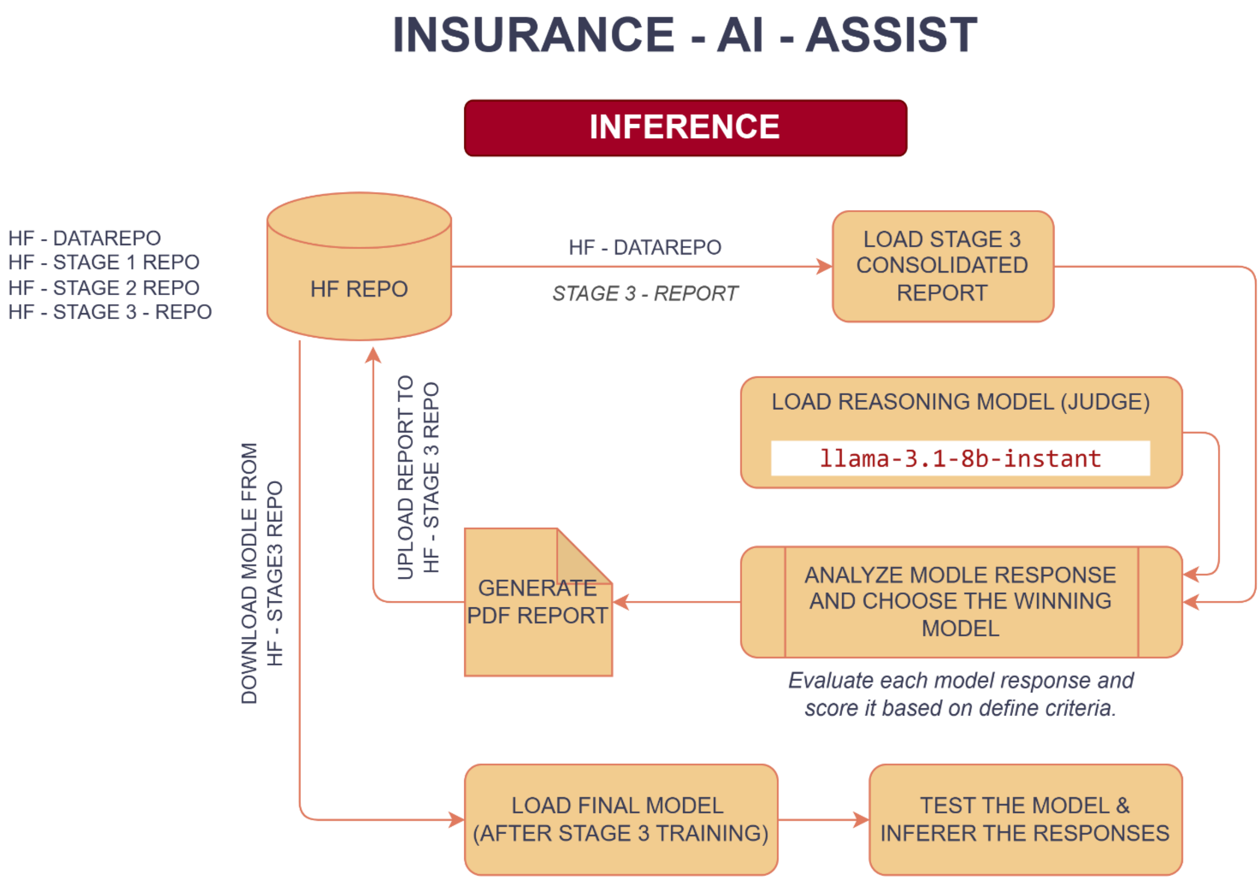
## STAGE 2



# INSURANCE - AI - ASSIST

## STAGE 3






# 5. Dataset Details

**Dataset Type**

Custom Commercial Insurance Dataset (PDF & JSNOL)

**Data Sources**

The dataset was created using domain-specific insurance and underwriting content covering:

- Commercial insurance
- Risk assessment
- Workers compensation
- Property insurance
- Liability coverage
- Middle market underwriting
- Submission requirements

- Risk mitigation practices
- Insurance operations

**Dataset Transformation Stages**

**Stage 1 Dataset** – Non-instruction insurance knowledge dataset.

Format:

- Question
- Answer

**Stage 2 Dataset** – Instruction-following dataset.

Format:

- Instruction
- Input
- Response

**Stage 3 Dataset** – Preference dataset for DPO training.

Format:

- Prompt
- Chosen Response
- Rejected Response

# 6. Model Used

**Base Model –** Qwen2.5-1.5B-Instruct

The model was selected because:

- Lightweight and efficient for fine-tuning
- Strong instruction-following capabilities
- Suitable for QLoRA-based training
- Works effectively within Google Colab GPU limitations
- Provides a good balance between performance and computational cost

**Judge Model –** Llama-3.1-8B-Instant

The judge model was used during the evaluation phase to compare responses from:

- Base Model
- Non-Instruction Fine-Tuned Model
- Instruction Fine-Tuned Model
- DPO Fine-Tuned Model

The judge model assigned scores and identified the best response based on:

- Correctness
- Completeness
- Domain Accuracy
- Professional Quality
- Helpfulness

## 7. Non-Instruction Fine-Tuning Approach

**Objective**

Teach the model insurance domain knowledge.

**Method**

The model was trained on insurance-related question and answer pairs without explicit instruction formatting.

**Outcome**

The model learned:

- Insurance terminology
- Underwriting concepts
- Risk assessment language
- Commercial insurance knowledge

However, response structure and instruction-following capability remained limited.

## 8. Instruction Fine-Tuning Approach

**Objective**

Improve the model's ability to follow user instructions and provide structured responses.

**Method**

Training examples were converted into instruction-response format.

**Outcome**

The model became better at:

- Understanding user intent
- Providing structured answers
- Maintaining conversational format
- Producing professional responses

## 9. DPO Alignment Approach

**DPO**

**Direct Preference Optimization**

**Objective**

Improve answer quality by learning user preferences.

**Approach**

The model learns which response is preferred and adjusts its behavior accordingly.

**Outcome**

The model generated:

- More helpful answers
- Better clarity
- Better professionalism
- Reduced hallucinations
- Improved alignment with expected insurance responses

## 10. LoRA / QLoRA Configuration

QLoRA was selected because it enables efficient fine-tuning on limited GPU resources by loading the model in 4-bit precision while training LoRA adapters.

## 11. Training Screenshots or Logs

Refer below for training logs –

https://github.com/Pzazz55/insurance-ai-assistant-finetuning/tree/main/reports

## 12. Before vs After Output Comparison

Refer the below final evaluation report –

https://github.com/Pzazz55/insurance-ai-assistant-finetuning/blob/main/reports/final_evaluation_20260710_025732.pdf

## 13. Final Observations

**Key Findings**

- Domain adaptation significantly improved insurance knowledge.
- Instruction tuning improved response structure.
- DPO alignment improved response quality and professionalism.
- QLoRA enabled efficient training on limited hardware.
- Preference optimization produced the most consistent and useful outputs.

## 14. Challenges Faced

**Technical Challenges**

- Limited GPU memory in Google Colab
- Token length limitations
- Parsing large evaluation reports
- Managing multiple training stages
- DPO dataset creation and validation
- Consistent evaluation across model checkpoints

**Solutions**

- Used QLoRA for memory efficiency
- Applied truncation strategies
- Built automated evaluation pipelines
- Implemented model comparison frameworks

## 15. Future Improvements

**Planned Enhancements**

1. **Retrieval Augmented Generation (RAG)**

   Integrate:

   - Underwriting manuals
   - Policy documents
   - Regulatory references

2. **Insurance Knowledge Base**

   Create a searchable repository of insurance documents.

3. **Vector Database Integration**

   Implement a Vector Database like – FAISS, ChromaDB, Pinecone, etc.

4. **Multi-Agent Architecture**

   Develop specialized agents:

   - Underwriting Agent
   - Claims Agent
   - Risk Assessment Agent
   - Compliance Agent

5. **Web Application**

   Convert the solution into an interactive application using:

6. **Continuous Preference Learning**

   Expand DPO datasets using production feedback and user interactions.

## 16. Important Links

- GitHub – https://github.com/Pzazz55/insurance-ai-assistant-finetuning/tree/main
- HuggingFace – https://huggingface.co/Pzazz55

## Conclusion

This project successfully developed a domain-adapted Insurance AI Assistant through a three-stage fine-tuning pipeline consisting of:

1. Non-Instruction Fine-Tuning

2. Instruction Fine-Tuning
3. DPO Preference Alignment

Using QLoRA enabled efficient training on limited GPU resources while achieving strong performance in commercial insurance and underwriting tasks. The final DPO-aligned model demonstrated the highest quality, professionalism, and user alignment among all evaluated model versions.